

TEST_CASES

NL-TO-SQL ANALYTICS AGENT

1. SQL GENERATION TESTS

Test Case ID	Input Query	Expected Result	Status
TC001	Total revenue by category	Valid <code>SELECT</code> query generated with <code>GROUP BY</code>	PASS
TC002	Show monthly sales trend	Valid <code>SELECT</code> query with date grouping	PASS
TC003	Top 5 customers by revenue	Valid <code>SELECT</code> query with <code>ORDER BY DESC LIMIT 5</code>	PASS
TC004	Highest selling product	Valid <code>SELECT</code> query with <code>ORDER BY DESC LIMIT 1</code>	PASS

2. SQL VALIDATION TESTS

Test Case ID	Input Query	Expected Result	Status
TC101	DROP TABLE users	Query blocked	PASS
TC102	DELETE FROM customers	Query blocked	PASS
TC103	UPDATE sales SET price=0	Query blocked	PASS
TC104	INSERT INTO orders VALUES(...)	Query blocked	PASS
TC105	WITH cte AS (...) SELECT * FROM cte	Query allowed	PASS

3. DATASET UPLOAD TESTS

Test Case ID	Input	Expected Result	Status
TC201	Upload retail_sales_dataset.csv	Dataset loaded successfully	PASS
TC202	Upload SQLite database	Database loaded successfully	PASS
TC203	Upload unsupported file type	Error message displayed	PASS

4. QUERY EXECUTION TESTS

Test Case ID	Input	Expected Result	Status
TC301	Execute valid SELECT query	Results displayed correctly	PASS
TC302	Query returns no rows	Empty result handled gracefully	PASS
TC303	Query with aggregate functions	Correct calculations displayed	PASS

5. CHART GENERATION TESTS

Test Case ID	Input	Expected Result	Status
TC401	Category analysis	Bar chart generated	PASS
TC402	Monthly sales trend	Line chart generated	PASS
TC403	Product distribution	Pie chart generated	PASS
TC404	Revenue vs Quantity	Scatter chart generated	PASS
TC405	Download chart	PNG image downloaded	PASS

6. BUSINESS INSIGHT TESTS

Test Case ID	Input	Expected Result	Status
TC501	Revenue by category	Business insight generated	PASS
TC502	Sales trend analysis	Trend explanation generated	PASS
TC503	Top products analysis	Top performer insight generated	PASS

8. ERROR HANDLING TESTS

Test Case ID	Input	Expected Result	Status
TC701	Ollama unavailable	Fallback engine activated	PASS
TC702	No dataset uploaded	Warning message displayed	PASS
TC703	Invalid SQL query	Validation error displayed	PASS

9. INTEGRATION TESTS

Test Case ID	Input	Expected Result	Status
TC801	Upload → Query → Execute → Visualize	Complete workflow successful	PASS
TC802	Multiple user queries	Session maintained correctly	PASS

TEST SUMMARY

Total Test Cases: 24

Status: PASS

Tested Modules:

- * Schema Tool
- * SQL Generator
- * SQL Validator
- * SQL Executor
- * Chart Generator
- * Insight Generator
- * Streamlit UI